POWER SUPPLY PROJECT CALCULATIONS

Prepared & Designed by Ali Hassan

Calculation for transformer :-

As $\frac{N_s}{N_p} = \frac{V_s}{V_p}$ As we know $V_p = 220v$

$V_s = 12v$

$$\frac{N_s}{N_p} = \frac{12}{220}$$

$$\boxed{\frac{N_s}{N_p} = \frac{3}{55}}$$

Calculation before diode (Bridge Rectifier)

As we know $V_s = V_{rms}$

So $\boxed{V_{rms} = 12v}$

As $V_{rms} = \frac{V_{max}}{\sqrt{2}}$

So $V_{max} = V_{rms} \times \sqrt{2}$

$= 12 \times \sqrt{2}$

$\boxed{V_{max} = 16.97v}$

$$V_{dc} = \frac{2\ V_{max}}{\pi}$$

$$= \frac{2 \times 16.97}{(3.14)}$$

$$\boxed{V_{dc} = 10.81v}$$

Calculation after bridge rectifier

Diode we use $\boxed{1N5408}$

Rating:- Peak Reverse Voltage = 1000v

Maximum RMS Voltage = 700v

Forward Current = 3A

Voltage drop across each diode ≅ 0.7 v

So Vmax after diode = $V_{max} - 2 \times 0.7$

$= 16.97 - 2 \times 0.7$

$= 16.97 - 1.4$

$$\boxed{V_{max} \text{ after diode } = 15.57v}$$

Ripple factor :-

$$\gamma = \frac{V_{r,rms}}{V_{dc}}$$

Finding V_{rms} & V_{dc} for V_{max} 15.57v :-

$$V_{rms} = \frac{V_{max}}{\sqrt{2}}$$

$$V_{rms} = \frac{15.57}{\sqrt{2}}$$

$$\boxed{V_{rms} = 11.01v}$$

$$V_{dc} = \frac{2 V_{max}}{\pi}$$

$$= \frac{2 \times 15.57}{(3.14)}$$

$$\boxed{V_{dc} = 9.92v}$$

$$V_{r,rms} = \sqrt{V_{rms}^2 - V_{dc}^2}$$
$$= \sqrt{11.01^2 - 9.92^2}$$
$$V_{r,rms} = 4.776$$

$$\gamma = \frac{V_{r,rms}}{V_{dc}}$$
$$\gamma = \frac{4.776}{9.92}$$
$$\gamma = 0.4814$$
$$\boxed{\gamma = 48.1\%}$$

Diode Rating:- {Diode we can use}

$$PIV = V_{max}$$
$$= 16.97$$

$$PIV \text{ rating} \geq 1.5 \times 16.97$$
$$\geq 25.4V \text{ (min)}$$

$$PIV \text{ rating} \geq 2 \times 16.97$$
$$\geq 33.94 \text{ (better)}$$

$$I_{d,max} = \frac{V_{max}}{R_L} = \frac{16.97}{680}$$
$$I_{d,max} = 24.9mA$$

$$I_{d,max} \text{ rating} \geq 1.5 \times 24.9$$
$$\geq 34.35mA \text{ (min)}$$

$$I_{d,max} \text{ rating} \geq 2 \times 24.9$$
$$\geq 49.8mA \text{ (better)}$$

Rating we need:

$PIV = 25.4 - 33.94v$

$I_{d,max} = 34.35 - 49.8mA$

Rating we have:

$PIV = 1000v$

$I_{d,max} = 3A$

Filter (Capacitor)

Capacitor we need for 5% ripple factor

$$C = \frac{1}{4\sqrt{3} \times r \times R_L \times f}$$

$$= \frac{1}{4\sqrt{3} \times 0.05 \times 680 \times 50}$$

$$\boxed{C = 85\,\mu F}$$

But as we have $4700\,\mu F$ capacitor so its ripple factor will be:-

$$\gamma = \frac{1}{4\sqrt{3} \times C \times R_L \times f}$$

$$= \frac{1}{4\sqrt{3} \times 4700 \times 680 \times 50}$$

$$\gamma = 0.0009$$

$$\boxed{\gamma = 0.09\,\%}$$

Calculation for Zener Diode:

Zener Diode we use is $\boxed{\text{BZV55-C12}}$

Rating from Data Sheet:-

Zener Voltage = $V_z = 12V$

Zener Voltage tolerance = 5%

Power dissipation = $P_{total} = 500mW$

$V_{z\,min} - V_{z\,max} = 11.40 - 12.70$

Test current = $I_{zT} = 5mA$

At 5mA Zener Max Resistance = $Z_z = 25\Omega$

Max Zener Current = $I_{zM} = 42mA$

Forward voltage = $V_{f\,(max)} = 0.9V$

Forward current = $I_{f\,(max)} = 250mA$

$V_{in(min)}$ & $V_{in(max)}$:-

At $I_{zk} = 5mA$

$$V_{out} = V_z - \Delta V_z$$

$$= V_z - (I_z - I_{zk}) Z_z$$

$$= 12V - (5mA - 5mA)\, 25\Omega$$

$$V_{out} = 12V$$

$$V_{in(min)} = I_{zk} R + V_{out}$$

$$= (5mA)(270\Omega) + 12V$$

$$\boxed{V_{in(min)} = 13.35\,V}$$

At $I_{zM} = 42mA$

$$V_{out} = V_z + \Delta V_z$$
$$= V_z + (I_{zm} - I_z)\, Z_z$$
$$= 12V + (42mA - 5mA)\, 25\Omega$$
$$V_{out} = 12.92V$$

$$V_{in(max)} = I_{zM} R + V_{out}$$
$$= (42mA)(270\Omega) + 12.92V$$

$$\boxed{V_{in(max)} = 24.26V}$$

R_{L} (min):-

$$R_{L(min)} = \frac{V_{out}}{I_{L(max)}}$$

$$I_{L(max)} = I_{z(max)} - I_{z(min)}$$

$$I_{z(max)} = \frac{V_{in(min)} - V_z}{R_z} = \frac{13.35-12}{270}$$

$$=$$

$$I_{z(max)} = \frac{V_{in(max)} - V_z}{R_z} = \frac{24.26-12}{270}$$

$$I_{z(max)} = 42mA$$

$$I_L = 42mA - 5mA \Rightarrow I_{L(max)} = 37mA$$

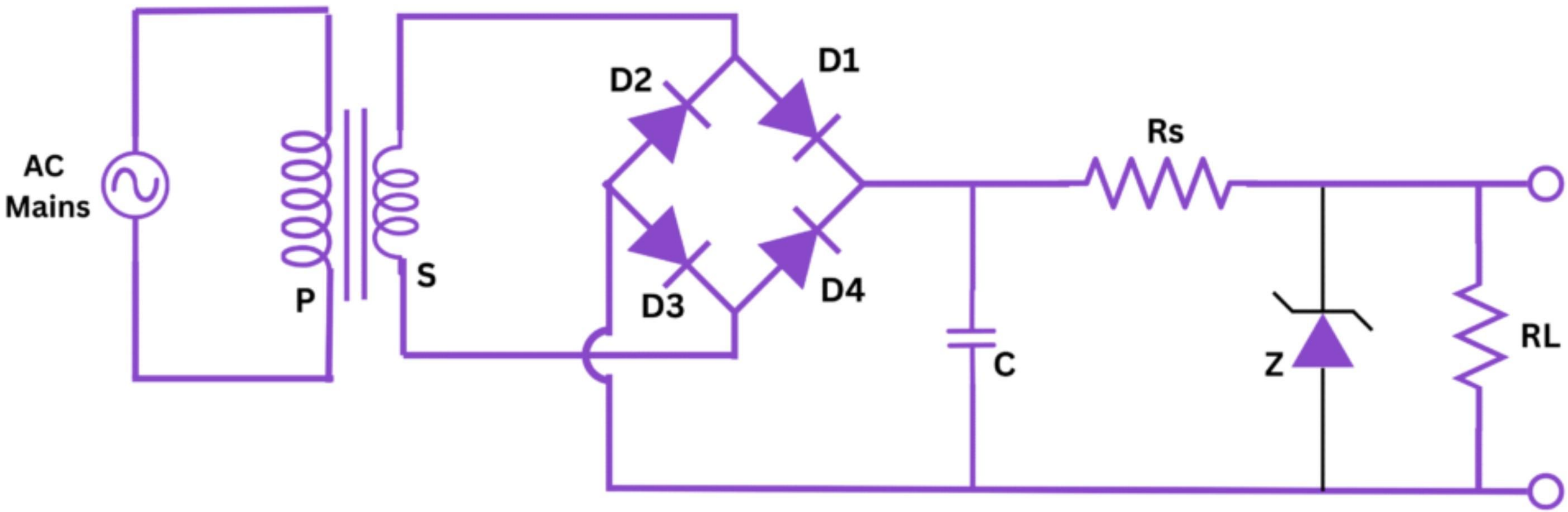
$$R_{L(min)} = \frac{12}{37mA}$$

$$\boxed{R_{L(min)} = 324\Omega}$$

Power dissipation:

$$P_{z(max)} = V_z \times I_z$$
$$= 12 \times 0.042$$

$$\boxed{P_{z(max)} = 0.5W}$$



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